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Treatment of chronic left ventricular assist device infection with local application of bacteriophages

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Abstract

Left ventricular assist device-associated infections represent a major complication during long-term support. Driveline exit site infections harbour the risk of ascending into the pump, causing deep-seated infections. We report on the successful treatment of a chronic recurrent left ventricular assist device pump-associated infection by the local application of bacteriophages as adjunct to standard surgical and antimicrobial treatment.

Keywords: Left ventricular assist device-associated infections • Methicillin-susceptible *Staphylococcus aureus* • Bacteriophages

CASE REPORT

A 67-year-old patient with ischaemic cardiomyopathy and a history of mechanical mitral valve replacement by median sternotomy underwent implantation of a left ventricular assist device (LVAD) (HeartWare HVAD, Framingham, MA, USA) as destination therapy via a left lateral thoracotomy. Four days after implantation, the pump had to be exchanged due to pump thrombosis. Three weeks later the patient was discharged with primary healing of the wound.

After 4 years on support, the patient developed an abscess at the thoracotomy scar site with involvement of the pump. Surgical debridement and vacuum-assisted closure (VAC) therapy were initiated. Intraoperative culture samples tested positive for methicillin-susceptible *Staphylococcus aureus*. Intravenous therapy with piperacillin/tazobactam was initiated and the wound was closed 14 days later, after 2 further VAC exchanges. However, 2 weeks later the patient developed a recurrent infection with purulent wound discharge. He was readmitted to the hospital and wound debridement with partial resection of the 6th rib was performed to gain better access to the abscess cavity (Fig. 1). VAC treatment was again combined with intravenous antibiotic therapy with piperacillin/tazobactam. After 31 days, the thoracic wall was reconstructed with a latissimus dorsi flap and the patient was discharged after a total of 6 weeks in hospital.

Another 4 months later, the patient was readmitted to hospital with fever and elevated inflammatory markers. A computed

tomography scan was performed, which showed an abscess around the pump. The wound was reopened and VAC treatment reinitiated. Intraoperative culture samples tested positive for methicillin-susceptible *Staphylococcus aureus*. VAC therapy was continued for 29 days. The patient provided informed consent for bacteriophage treatment as a last-resort treatment. After the wound was cleaned and exhibited growth of granulation tissue, the skin was closed by secondary intention. During skin closure, bacteriophages active against multiple bacteria, including *Staphylococcus aureus*, were applied locally. The bacteriophage solution (10 ml) was a 1:1 mixture of PYO bacteriophage with $\sim 10^6$ plaque-forming units/ml (Georgian pharmaceutical product registration number: R-022600) and Staphylococcal bacteriophage Sb-1 with $\sim 10^7$ plaque-forming units/ml (R-022876), both obtained from the Eliava Institute of Bacteriophage, Microbiology and Virology, Tbilisi, GA. Bacteriophage products are a mixture of sterile filtrates of phage lysates which have previously been shown to be active against *S. aureus* isolates from the patient. A drain was placed into the surgical site through which bacteriophages (5 ml) were applied every 8 h for 10 days. After surgery, intravenous antibiotics (meropenem and fosfomycin) were administered, followed by oral antibiotics (ciprofloxacin/ri-fampicin). Except for mild nausea, no adverse effects were observed. No antiseptics were used during surgery. The wound healed and the patient was discharged after 16 days. At the follow-up visit 9 months later, the surgical site showed no local signs of infection (Fig. 2).

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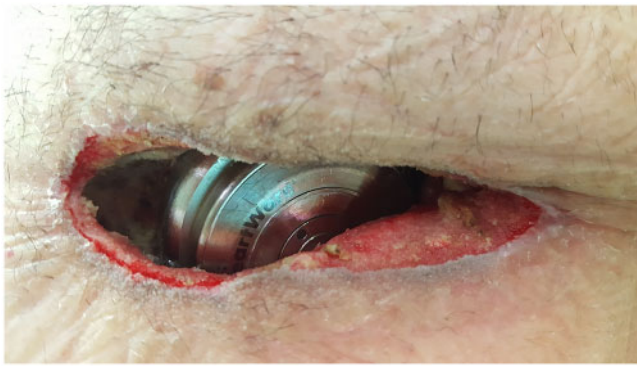


Figure 1: Infected pump during VAC exchange.

DISCUSSION

LVAD-associated infections represent a serious and potentially life-threatening complication, and are associated with substantial morbidity and mortality. Additionally, during prolonged antibiotic treatment, antimicrobial resistance and adverse effects may develop. Therefore, alternative anti-bacterial approaches were evaluated.

Following failure of antibiotic therapy, surgical debridement and exchange of device parts, the recommended treatment of recurrent driveline or pump infections includes VAC therapy and wrapping with mobilized greater omentum or latissimus dorsi muscle [1]. In the case presented here, there was close contact between the thoracic wall and the LVAD pump. Strong adhesions precluded transposition of the omentum, therefore, the latissimus dorsi muscle was mobilized to cover the pump. However, 4 months later the infection returned. One factor that discouraged primary pump exchange was the need to implant the new pump into an infected area.

Initial experience with bacteriophage therapy was gained more than 100 years ago [2]. Bacteriophages are viruses that target and infect specific host bacteria, replicate inside them and then destroy them [3]. Moreover, phages are able to penetrate into microorganisms in biofilms on implanted parts of the LVAD. In the described case, bacteriophages cleaned the wound, enabled successful skin closure and resulted in an infection-free period of 9 months thus far.

Though the experience with bacteriophages is limited in Western countries [4], this treatment is commonly used in clinical practice in East European countries that have access to commercial bacteriophage products. Phages can be administered subcutaneously, intramuscularly, intravenously, orally, topically or through inhalation. A clinical, double-blind placebo-controlled trial in the UK showed physical improvement in patients with chronic *Pseudomonas aeruginosa* otitis media, whereas a randomized trial with a topically applied phage cocktail in burn



Figure 2: Five months after topical application of bacteriophages and secondary wound closure.

wounds has also been published. Early clinical experience with bacteriophage therapy in 3 lung transplant recipients and in a cystic fibrosis patient has shown promising results [5; see also [Supplementary Material](#)].

Our experience justifies further investigation into the indications for and the therapeutic potential of bacteriophage use in LVAD-associated infections.

SUPPLEMENTARY MATERIAL

[Supplementary material](#) is available at *EJCTS* online.

Conflict of interest: none declared.

REFERENCES

- [1] Potapov EV, Antonides C, Crespo-Leiro MG, Combes A, Färber G, Hannan MM et al. 2019 EACTS Expert Consensus on long-term mechanical circulatory support. *Eur J Cardiothorac Surg* 2019;56:230–70.
- [2] Chanishvili N. Phage therapy—history from Twort and d'Herelle through Soviet experience to current approaches. *Adv Virus Res* 2012; 83:3–40.
- [3] Wienhold MS, Lienau J, Witzernath M. Toward inhaled phage therapy in Western Europe. *Viruses* 2019;11:pii:295. doi:10.3390/v11030295.
- [4] Aslam S, Pretorius V, Lehman SM, Morales S, Schooley RT. Novel bacteriophage therapy for treatment of left ventricular assist device infection. *J Heart Lung Transplant* 2019;38:475–6.
- [5] Law N, Logan C, Yung G, Furr CL, Lehman SM, Morales S et al. Successful adjunctive use of bacteriophage therapy for treatment of multidrug-resistant *Pseudomonas aeruginosa* infection in a cystic fibrosis patient. *Infection* 2019;47:665–8.